

Habitat imaging analysis of local failures in skull-base cancer patients treated with proton therapy

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Abstract

Background and Motivation: Accurate modelling of treatment response in proton therapy is challenging due to inter-patient variability and tumour heterogeneity, which can lead to local tumour control failure. While quantitative imaging biomarkers hold promise for predicting outcomes, standard radiomics approaches often fail to account for spatial intra-tumoural heterogeneity, which is increasingly recognized as important for modelling tissue-level effects. To address this, habitat imaging partitions tumours into spatially distinct subregions (habitats) with homogeneous multimodal imaging properties. This approach aims to better capture the spatial distribution and prevalence of complex biological phenotypes for predictive modelling. In this project, it is applied to survival analysis and the prediction of local failure after proton therapy. *Materials and Methods:* Habitat imaging analysis was performed in a cohort of 38 skull-base cancer patients treated with proton therapy, of whom 7 experienced tumour recurrence, defined as local failure. Baseline multi-modal imaging consisting of CT and MRI sequences including T1, FLAIR, DWI/ADC and contrast-enhanced GADO-T1, was collected consistently across the cohort. After pre-processing, the soft-tissue component of the gross tumour volumes (GTV) was analysed using six algorithms for clustering of MR image intensities into high- and low-intensity sub-regions, yielding 16 habitats. Clustering alternatives went from standard K-Means to advanced spatially aware techniques. The prognostic relevance of individual habitat volume fractions and their presence within the GTV was assessed using a bootstrap procedure with an L1-penalized Cox proportional hazards model. Decisive factors included selection frequency, consistent weight sign and model concordance index (C-index). *Results:* Although individual habitat volume fractions did not reach statistical significance in corrected univariate tests, multivariate survival models demonstrated predictive performance for local failures. K-Means and Otsu clustering algorithms yielded the highest concordance indices, outperforming spatially aware algorithms. Specific habitats were consistently selected across clustering methods: the habitat with high GADO-T1 intensities acted as a robust hazard-increasing predictor, whereas habitats with hyperintense FLAIR or combined high ADC and FLAIR consistently demonstrated hazard-decreasing effects. *Conclusion:* A structured approach for generating and analysing tumour habitats from multi-modal MR images was developed. The identified habitats highlight the potential of spatial intra-tumour heterogeneity to serve as a predictive signature for local failure. External validation in larger cohorts and combination with dosimetric data are necessary to translate these findings into personalized treatment optimizations in a clinical context.

Contents

1	Introduction	3
1.1	Proton Therapy	3
1.2	Outcome Modelling in Radiotherapy	3
1.3	Tumour Heterogeneity	5
1.4	Imaging for Outcome Modelling	6
1.5	Habitats	7
1.6	Study Goals	7
2	Materials and Methods	8
2.1	Clinical Cohort	8
2.2	Habitat Development Pipeline	9
2.3	Feature Extraction	15
2.4	Survival Analysis	16
3	Results	21
3.1	GTV Habitat Distributions	21
3.2	Cox Proportional Hazards Model Results	21
4	Discussion	27
4.1	Interpretation of Results	27
4.2	Limitations	30
4.3	Uncertainty	31
4.4	Future Work	32
5	Conclusion	32
	Appendix	37
A	Habitat Volume Fraction Statistics	37
B	Co-Selection Habitat Network	40

1 Introduction

1.1 Proton Therapy

Proton therapy is an advanced form of external beam radiotherapy that uses accelerated protons instead of conventional photons to treat tumours with high precision. A key physical advantage of protons is their finite range in tissue, characterized by the Bragg peak, which allows high dose deposition near the end of their range, followed by a rapid dose fall-off beyond the target volume [1]. This physical property enables improved sparing of healthy tissue surrounding deep-seated tumour masses compared with photon-based radiotherapy, particularly in radiosensitive regions such as the brain.

Despite these dosimetric advantages, clinical outcomes after proton therapy remain influenced by inter-patient variability in treatment response, which may be partly related to tumour heterogeneity, and can result in failure of tumour control. Therefore, accurate modelling of treatment response and failure risk beyond dosimetric analysis alone could provide quantitative support for treatment optimisation. Patient-specific prediction of the therapeutic response to therapy could inform clinical decision-making when selecting among treatment options and evaluating potential risks and benefits, ultimately promoting the development of personalized radiotherapy paradigms [2].

1.2 Outcome Modelling in Radiotherapy

Tumour response to radiotherapy is commonly modelled using tumour control probability (TCP) models, which in their most simple formulations estimate the likelihood of achieving tumour eradication based primarily on the treatment dose and fractionation scheme. These models are generally modality-specific and often derived from population-level statistics rather than individualized patient characteristics. An alternative, but complementary, approach is statistical survival modelling (time-to-event analysis), which incorporate a temporal outlook, but also allows extension to patient-specific variables to refine risk estimates on an individual basis.

Advances in quantitative imaging have enabled the extraction of imaging biomarkers and radiomic features that characterize tumour phenotypes beyond visual assessment. These quantitative imaging biomarkers (QIBs), derived from modalities such as CT and MRI, provide standardized and reproducible descriptors of tumour heterogeneity and tissue properties, and are investigated as complementary predictors of treatment response and treatment outcome.

TCP models

TCP models are usually based on radiobiological principles, describing cell survival following radiation, and aim to quantify the probability that irradiation successfully eliminates all cancer cells capable of continued proliferation. These models are derived from Poisson-based assumptions describing the stochastic survival of clonogenic cells after treatment [3].

Given n number of clonogenic cells, tumour control is achieved when no surviving clonogens remain ($n = 0$) after irradiation. The corresponding TCP can therefore be written as

$$TCP = P(n = 0) = e^{(-n_0 \cdot S(D))}, \quad (1)$$

with n_0 representing the initial clonogenic cells and $S(D)$ describing the fraction of cells surviving after receiving a delivered dose D . In most radiobiological applications, the survival $S(D)$ is estimated using the linear-quadratic (LQ) model,

$$S(D) = e^{-\alpha D - \beta G D^2}, \quad (2)$$

where α and β are radiobiological parameters describing cellular radiosensitivity, and G is the protraction factor that accounts for intra-fraction cellular repair [4]. The parameter α describes the component of radiation-induced cell killing that increases linearly with dose, commonly associated with irreparable damage caused by single radiation events, while β represents a quadratic component that becomes more relevant at higher doses and reflects damage that results from the accumulation and interaction of sublethal lesions produced by multiple radiation tracks. The relative contribution of these two components is characterized by the α/β ratio, defined as the dose at which the linear and quadratic contributions to cell killing are equal [3, 4]. Tissues with a high α/β ratio, such as many tumours and early-responding tissues, are dominated by the linear component and therefore tend to be less sensitive to changes in dose per fraction. In contrast, tissues with a low α/β ratio, such as many late-responding normal tissues, exhibit a stronger quadratic contribution and are consequently more sensitive to fraction size; they are therefore relatively spared by fractionating treatment into smaller daily doses, allowing sublethal damage repair between fractions.

Although TCP models provide an interpretable framework for estimating tumour response to radiotherapy, they are typically based on population-averaged radiobiological parameters, such as α and β , assigned uniformly to entire tumour or tissue types. Consequently, they do not adequately account for intra-tumour heterogeneity or patient-specific biological variation that may influence treatment response. Furthermore, conventional TCP models generally predict tumour control endpoint, but do not incorporate information regarding the timing of recurrence or progression. In contrast, statistical survival models can provide a more flexible framework for relating patient-specific characteristics not only to the occurrence of an event, but also to the time until that event occurs.

Time-to-event statistical survival modelling

Statistical survival analysis provides a data-driven approach to evaluating clinical outcomes from a population-level perspective without relying on explicit radiobiological

assumptions. Rather than estimating cancer stem cell survival as a function of dose, survival models focus on the time until a clinically relevant event occurs, such as local failure, disease progression, or death. This allows treatment outcomes to be analyzed across heterogeneous patient cohorts while incorporating available patient-specific information. A key aspect of survival analysis is its ability to handle censored observations, which are common in longitudinal clinical studies. Censoring occurs when a patient does not experience the event of interest during the follow-up period, or when follow-up information becomes unavailable. By accounting for censored data, survival models can make use of incomplete observations and reduce bias, provided that censoring is appropriately modelled and is not strongly informative.

One of the most commonly used methods for survival modelling is the Cox proportional hazards (CPH) model. The CPH model relates a set of covariates to the hazard function (Eq. 3), which describes the instantaneous risk of experiencing the event at a given time. Covariates may include patient characteristics, treatment-related variables, or, as in the present study, quantitative imaging features. The CPH model can be expressed as

$$h(t|x) = h_0(t)e^{\beta^T x}, \quad (3)$$

where $h(t|x)$ represents the hazard at time t for a patient with covariate vector x , $h_0(t)$ is the baseline hazard, and β is the vector of regression coefficients associated with the covariates, also referred to as Cox coefficients. The effect of each covariate is commonly interpreted through the exponential of its coefficient, known as the hazard ratio (HR). Negative coefficients ($\text{HR} < 1$) represent hazard-decreasing covariates while positive coefficient ($\text{HR} > 1$) indicate an increase in hazard.

Due to their multivariable formulation, survival models are particularly well suited for integrating imaging-derived biomarkers alongside conventional dosimetric and clinical covariates, enabling more individualized modelling of treatment outcome. The predictive performance of survival models is commonly evaluated using the concordance index, or C-index, which measures the agreement between predicted risk scores and observed outcomes. Specifically, it quantifies the ability of the model to assign higher risk to patients who experience the event earlier. A value of 0.5 corresponds to random prediction, while a value of 1.0 indicates perfect concordance between predicted and observed outcomes.

1.3 Tumour Heterogeneity

Tumour heterogeneity refers to the presence of spatial and temporal variations in cellular, genetic, and microenvironmental properties within a tumour. These variations may reflect differences in blood perfusion, hypoxia, cellularity, and clonal composition, and can result in heterogeneous treatment sensitivity across different tumour regions [5].

In the context of radiotherapy, intra-tumoural heterogeneity is particularly relevant because it may influence local radioresistance and spatial patterns of recurrence. Regions with distinct biological properties may respond differently to radiation dose, leading to

non-uniform treatment response within the same tumour volume [6], and being a potential root cause for local failure. Capturing and quantifying this heterogeneity is therefore crucial information for predictive modelling and for developing biologically-driven treatment strategies.

1.4 Imaging for Outcome Modelling

Medical imaging provides a non-invasive means of characterizing tumour structure and heterogeneity. In radiotherapy, imaging plays a central role not only in treatment planning, patient positioning, and treatment verification, but also in the extraction of features that may be associated with clinical outcomes. Patient images can be converted into quantitative imaging biomarkers that can reflect underlying biological processes and may improve the predictive performance of outcome models.

Imaging modalities

Multiple imaging modalities are commonly used in proton therapy. Computed tomography (CT), which provides electron density information, is routinely used for treatment planning and dose calculation [7]. Magnetic resonance imaging (MRI), on the other hand, offers excellent soft-tissue contrast without additional ionizing radiation, and is increasingly included in pre-treatment imaging protocols for tumour delineation and staging.

In this study, different MRI sequences capturing distinct anatomical and physiological characteristics were used. Specifically, T1-weighted (T1) images provide structural anatomical information, while gadolinium-enhanced T1-weighted (GADO-T1) images highlight regions with contrast enhancement. Fluid-attenuated inversion recovery (FLAIR) is a T2-weighted imaging that suppresses cerebrospinal fluid signal, improving the visibility of lesions and oedema-like abnormalities in brain tissue [8]. Diffusion-weighted imaging (DWI) provides functional information related to water mobility, tissue cellularity, and diffusion restriction by measuring the microscopic motion of water molecules within tissue. From DWI acquisitions, apparent diffusion coefficient (ADC) maps can be derived, providing a quantitative representation of water diffusion. ADC maps are therefore frequently used in survival analysis models as imaging biomarkers related to tumour cellularity, with lower ADC values often associated with higher cellularity or specific states of cell replication [9].

The integration of multi-modal imaging enables a more comprehensive description of tumour phenotype than any single modality alone, making it particularly suitable for outcome modelling analyses. However, despite the complementary information provided by different imaging modalities, integrating these image-derived characteristics into robust biomarkers that capture multiple tumour properties remains challenging. Therefore, the development of habitat-based approaches aims to integrate multi-modal imaging information into spatially distinct tumour subregions.

1.5 Habitats

Outcome modelling based on imaging biomarkers often relies on extracted radiomics features, which typically summarises quantitative imaging features over the entire region of interest. As a result, structures such as the gross tumour volume (GTV) may be treated as homogeneous tissues. Although attempts to generate voxel-level radiomics maps have been reported in the literature [10], conventional radiomics approaches largely ignore intra-tumoural heterogeneity. Radiomics texture features, such as Gray Level Co-occurrence Matrix (GLCM), Size Zone Matrix (GLSZM), and Run Length Matrix (GLRLM), attempt to capture tissue imaging heterogeneity but suffer from a high sensitivity to image acquisition settings and preprocessing parameters, leading to poor reproducibility and generalizability. Habitat generation addresses these limitations by partitioning tumours into spatially distinct subregions, or “habitats”, without strong, absolute threshold definitions. Each habitat is intended to represent a region with relatively homogeneous properties derived from multimodal imaging, which may correspond to distinct biological phenotypes.

Habitats are typically derived by voxel-wise clustering of imaging intensities within a region of interest, in this case the tumour volume, using information from multiple modalities. This allows the construction of multi-parametric or multi-modal habitat maps that encode combinations of imaging characteristics across MRI sequences, thereby integrating complementary anatomical and functional information. Such representations provide a structured way to study spatial heterogeneity and, in this case, its relationship with treatment outcome.

However, similar to radiomics, habitat construction is not free from standardisation issues. It is sensitive to methodological choices, including image normalization, clustering strategy, noise and artifacts, and the use of spatial or intensity constraints. These factors can influence the stability of the resulting habitats, as well as subsequent intra-tumoural feature extraction and survival analysis.

1.6 Study Goals

The primary goal of this study was to assess the significance of habitat imaging analysis for predicting local in-field failure after proton therapy. The study addresses the methodological steps required to generate multimodal imaging habitats from baseline MR images acquired before proton therapy, and evaluates their use in survival analysis of local failure in skull-base chordoma.

A consolidated imaging framework available at the host institute, used to map functional and anatomical images onto a common anatomy, was extended with tools for habitat construction across multiple imaging modalities, including several clustering strategies. Different clustering methods for habitat definition were compared, including intensity-based and spatially aware alternatives, and their impact on predictive performance was assessed.

Finally, habitat features, including their volume fractions and prevalence in the primary tumour at baseline imaging, were used in Cox proportional hazards models to evaluate their predictive value for local failure after treatment. Through this approach, a set of potentially relevant habitats, defined as robust imaging-defined tumour subregions, was identified as a candidate signature associated with clinical failure.

2 Materials and Methods

2.1 Clinical Cohort

The initial clinical cohort consisted of 58 patients diagnosed with chordomas or chondrosarcomas who were treated with proton therapy at the Paul Scherrer Institute (PSI). All patients were treated between 2015 and 2023 with in-house-developed machines gantry 1 or gantry 2. From this initial cohort, only patients with baseline imaging performed at PSI were considered for analysis. This ensured consistency of image acquisition and avoided inter-center variations of imaging devices. Furthermore, patients with incomplete datasets, missing MRI sequences or insufficient quality for registration were excluded. Hence, the final study cohort consisted of 38 patients (65% of the initial clinical cohort) with 7 patients (18% of final study cohort) experiencing local failure post-treatment. All selected patients had complete baseline multimodal image records, consisting of computed tomography (CT) and four magnetic resonance imaging (MRI) sequences: diffusion-weighted imaging (DWI), T1-weighted imaging (T1), fluid-attenuated inversion recovery (FLAIR), and gadolinium-enhanced T1-weighted (GADO-T1) sequences. The patient selection process is illustrated in the PRISMA-style diagram shown in Fig. 1.

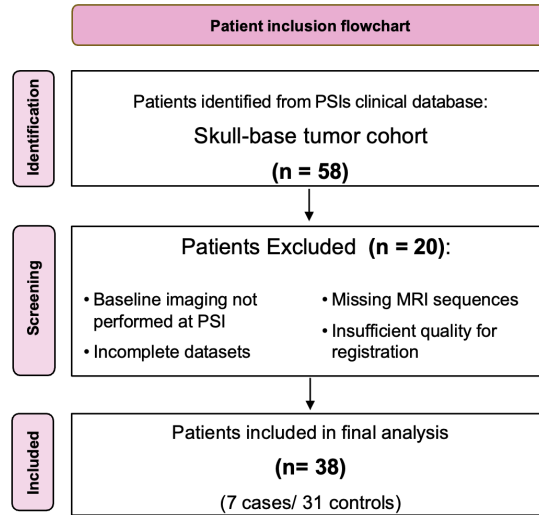


Figure 1: Prisma diagram visualizing the identification and screening procedures for the patient selection of the final study cohort.

2.2 Habitat Development Pipeline

The habitat-creation process was based on the adaptation of an existing brain radiomics pipeline [11, 12]. The original workflow comprised image registration, region-of-interest masking, intensity normalization, and feature extraction. To enable habitat generation, two additional stages were developed: binary clustering of voxels within the tumour volumes based on binary clustering of MRI image intensities, and the combination of clustered images for habitat definition. The following sections describe each component of the pipeline and the modifications introduced.

Image registration

To ensure anatomical consistency between different MRI sequences, all images were spatially aligned to the treatment planning CT reference frame, where the GTVs were contoured. Anatomical-based images (T1, GADO-T1, and FLAIR) were directly registered to the CT, while DWI images (converted to ADC maps) required intermediate registration to T1-weighted images before being registered to the treatment planning CT. All registration procedures were rigid (6 degrees of freedom) and implemented in Python using the SimpleITK library [13]. Mutual information was used as the similarity optimization metric.

To perform the registration, binary brain and skull masks for CT, T1, GADO-T1, and FLAIR images were computed using TotalSegmentator [14] and further converted to signed distance fields (SDF). SDFs provide a smooth representation of the mask boundaries, ensuring a more robust boundary-based registration. Rigid registration was performed for the FLAIR, T1, and GADO-T1 SDFs as moving images into the fixed CT SDF to obtain initial registration transformations. Affine registration was then used, where the final T1-CT, GADO-T1-CT and FLAIR-CT transforms were obtained [11].

Since ADC maps display functional information, additional preprocessing was needed for registration. Intensity values were first restricted to 0.3×10^{-3} and 1.5×10^{-3} mm²/s to exclude background noise and include the majority of brain tissue. The values within this range were then linearly scaled to the interval [0, 1], yielding soft masks of ADC maps that were inverted and smoothed. The normalized intensities were then inverted to approximate the contrast characteristics of T1-weighted images, enabling the use of the same registration framework described above [11]. For the final ADC-CT transform, the ADC-T1 and T1-CT registration transforms were concatenated.

Soft tissue masks

Once all images were registered, CT contours of the gross tumour volume (GTV), clinical target volume (CTV), and planning target volume (PTV) were used to create their corresponding binary masks. To ensure that the images contained only voxels representing soft tissue, additional preprocessing was performed to remove air and bone structures. Indeed, although the GTV may include or border bony structures, this work focuses

exclusively on the soft-tissue component of the GTV. ADC maps do not provide meaningful information in bone regions due to the absence of water diffusion in these areas. Following the methodology proposed in previous work [11], air was removed from the masks by thresholding the CT images between -1024 and -500 HUs.

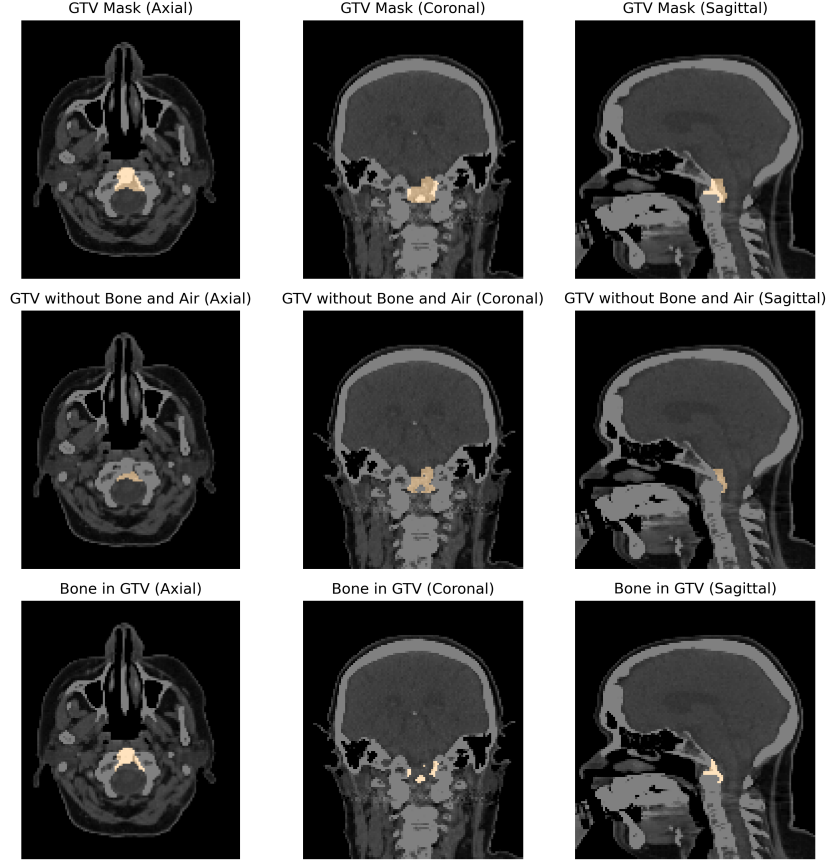


Figure 2: Comparison of the GTV mask, the GTV mask without bone and air, and the bone regions within the GTV mask. Air and bone removal was performed by thresholding and GMM-based clustering, respectively.

Bone removal was performed using a two-step procedure. First, the previously obtained skull binary mask (TotalSegmentator [14]) was subtracted from the regions of interest. Second, an additional refinement step introduced in this work employed a Gaussian Mixture Model (GMM)-based probabilistic clustering algorithm to separate the GTV into soft-tissue and bone, enabling a patient-wise bone-threshold definition (Fig. 2). The last step of the masking procedure was to obtain a white matter mask.

Image resampling and intensity normalization

To use the same binary masks on the registered MRI sequences required all images to have the same voxel spacing. To account for differences between voxel parameters, all images were resampled to match the CT voxel spacing.

Additionally, since MRI signal intensities do not express physical quantities (as opposed to CT) and are highly dependent on image acquisition parameters, image normalization across patients was necessary. For this purpose, normal appearing white matter (NAWM) was used. White matter provides a homogeneous, reference tissue in the brain and therefore the NAWM mask was used for normalization of the soft tissue for T1, GADO-T1, and FLAIR images. The masks were generated by first removing all structures except the brain, followed by the Automated Segmentation Tool (FAST) [15], a tissue classification algorithm from the FMRIB toolkit (FSL). For the ADC maps, white matter was not used, and the reference tissue for normalization was the brain without the GTV. Additionally, for the bone sections of the GTV, the reference tissue used was the bone mask excluding the GTV, though they were not further used for analysis.

Normalization was then performed using robust z-scoring. The median intensity, I_{median} , and median absolute deviation, MAD, of the reference tissue (white matter or brain without GTV) were calculated, and the normalized voxel values were obtained using Eq. 4 [11].

$$I_{\text{normalized}} = 0.6745 \cdot \frac{I_{\text{image}} - I_{\text{median}}}{\text{MAD}} \quad (4)$$

Data extraction

In order to use normalized GTV image intensities for habitat creation, said intensities were extracted from the MRI images. Initially, raw voxel intensities were obtained from each registered image, and background noise was smoothed using an isotropic 1.5mm standard deviation Gaussian filter, ensuring physically uniform smoothing while preserving transition regions and tissue barriers. These intensities were then normalized following Eq. 4 to ensure comparability between patients and modalities. Following normalization, the predefined masks were applied, and only normalized voxel intensities within the selected regions were retained for subsequent analysis.

Clustering

Once normalized voxel intensities within the GTV were obtained for all MRI sequences (Fig. 5b), the next step towards habitat formation involved partitioning each image into regions of high and low signal intensity. This discretization was performed to reduce the continuous intensity distributions while preserving intra-tumoural heterogeneity. Voxels were subsequently labelled by their high/low intensity combinations across the different imaging modalities, creating habitats. To achieve this, several clustering algorithms were explored and their respective habitat maps were later produced.

As a first approach and following existing habitat-imaging studies [16, 17, 18, 19], K-means clustering was applied. This clustering algorithm separates data into a predefined number of clusters by minimizing the variance within each cluster. In this work, the number of clusters was set to two ($k = 2$), corresponding to low and high intensity regions. The algorithm iteratively assigns each voxel intensity to one of the two clusters based on the distance to the cluster centroid, and updates the cluster centroids until convergence. It was implemented using the scikit-learn Python library, where the algorithm minimizes the objective function,

$$J = \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|^2, \quad (5)$$

where C_k represents the voxels assigned to cluster k and μ_k represents the cluster centroid. To ensure robustness, the clustering was repeated ten times with different initializations, and the solution with lowest variance within clusters was selected. The voxels were then labelled as high-valued (1) or low-valued (0) and mapped back to the original image space to generate a clustered GTV image for each MRI sequence.

As a second option, Otsu thresholding [20], also commonly used for intensity-based habitat generation [21, 22, 23, 24], was used. This algorithm is not iterative, but instead uses the distribution of intensities to determine an optimal cutoff threshold by maximizing the separation between two classes. The optimal threshold, t , is optimized by maximizing the between-class variance,

$$\sigma_b^2(t) = \omega_1(t)\omega_2(t) [\mu_1(t) - \mu_2(t)]^2, \quad (6)$$

where ω_1 and ω_2 represent the fraction of voxels within each of the two intensity classes and μ_1 and μ_2 are their corresponding mean intensities. Voxels with intensities below or equal to the computed threshold were labelled as high or low, and the clustered GTV images were generated.

Thirdly, a gaussian mixture model (GMM) [25] algorithm, also commonly used for habitat creation [26, 27, 28, 29], was also tested. In contrast to hard clustering methods, GMMs assume that voxel intensities are generated from a mixture of Gaussian distributions and assign each voxel a probabilistic chance to belong to each cluster. As with K-means, the number of clusters was set to two. In this algorithm, the means and covariances of each gaussian are estimated using the Expectation-Maximization algorithm [25], an iterative process that alternates between estimating the probability of each voxel belonging to each gaussian (expectation step) and updating the parameters of the Gaussian distributions based on these probabilities (maximization step). These steps are repeated until convergence and each voxel is then assigned to the cluster with the highest probability, resulting in in two discrete intensity groups, labelled as high and low. An example of the fitted Gaussian distributions over the GTV voxel intensity histogram is shown in Fig. 3.

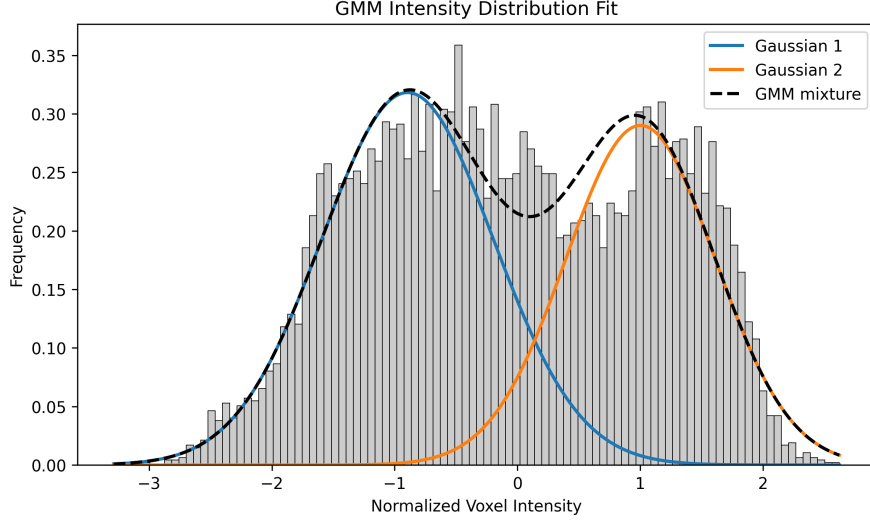


Figure 3: Gaussian mixture model fit of GTV voxel intensity distribution for a sample FLAIR MRI image.

While K-means, Otsu thresholding and standard GMM rely exclusively on voxel intensity values, they do not explicitly account for the spatial organization of voxels. As a result, the generated clusters may contain isolated voxels or fragmented regions that are unlikely to represent biologically meaningful information. To address this limitation, spatial information was incorporated into the clustering framework in two different ways.

As a first approach, a spatially-constrained Gaussian Mixture Model (spatial GMM) was implemented. In addition to voxel intensities, spatial coordinates (x, y, z) were included as features during clustering. Coordinates were both normalized by subtracting their mean, and divided by their standard deviation to achieve a uniform variance in all directions, avoiding directional bias. Additionally, a weight of 5% was applied to spatial coordinates contribution to GMM to ensure that the clusters were formed primarily by the voxel intensities. This approach encourages neighbouring voxels with similar intensities to be grouped together, promoting smoother and more spatially coherent clusters, while still preserving the probabilistic nature of the GMM framework.

Lastly, since the GMM algorithm provides voxel probabilities that reflect the uncertainty of voxels belonging to their selected cluster, a Markov Random Field (MRF)-inspired approach [30, 15], that uses such probabilities to enforce spatial coherence, was developed and applied to the standard GMM and spatial GMM images. The objective of this method was to further enforce local spatial consistency by encouraging each voxel to adopt the cluster most compatible with those of its neighbouring voxels. The algorithm was implemented using an Iterated Conditional Modes (ICM) optimization strategy [30, 15], in which voxels are iteratively updated by minimizing an energy function composed

of two terms: a data term and a spatial smoothness term (Eq. 7).

$$E = E_{\text{data}} + E_{\text{smooth}} \quad (7)$$

The data term (Eq. 8) was defined as the negative logarithm of the probability of the voxel belonging to the cluster, $P_c(x)$, hence favoring labels with higher probabilities.

$$E_{\text{data}} = -\log [P_c(x)]. \quad (8)$$

The smoothness term,

$$E_{\text{smooth}} = \lambda (N - n_x), \quad (9)$$

was the number of disagreeing neighbour voxels, where N was the total number of neighbours, n_x was the number of agreeing neighbours, and λ was set to 1. E_{smooth} penalized differences between neighbouring voxels using a 3-dimensional, six-connectivity kernel structure (face-adjacent neighbours of the center voxel), encouraging spatially homogeneous regions. During each iteration, the cluster associated with the minimum total energy was assigned to each voxel, and the process was repeated for three iterations. Both the standard GMM-MRF and spatial GMM-MRF clustered images were then saved.

After all clustering algorithms were applied to all patients, a total of six clustered images (K-means, Otsu, GMM, spatial GMM, GMM-MRF, and spatial GMM-MRF) were obtained, for each of the four MRI sequences (T1, FLAIR, ADC, GADO-T1). An example of all clusters produced from a FLAIR image of a sample patient is shown in Fig. 4 for the same image slice at the base of the skull.



Figure 4: Clustering results for a FLAIR image after performing six different types of clustering.

Habitat construction

Following the generation of clustered images for each MRI sequence and each clustering method, habitat maps were constructed to integrate information across all modalities. A binary encoding system was used, uniquely identifying every possible combination of high- and low-intensities across the four MRI sequences. For each voxel, a vector of binary labels was formed. Each MRI sequence contributed one bit, describing whether the voxel belonged to the high or low intensity region. Combining these bits generated a unique label for every possible habitat configuration, and each habitat was given a specific numerical value from 1 to 16. All possible combinations and habitat definitions are shown in Fig. 6. The resulting habitat map therefore represents all possible label

combinations in a single volumetric image. Fig. 5c shows an example of one slice of the habitat images created by different clustering techniques.

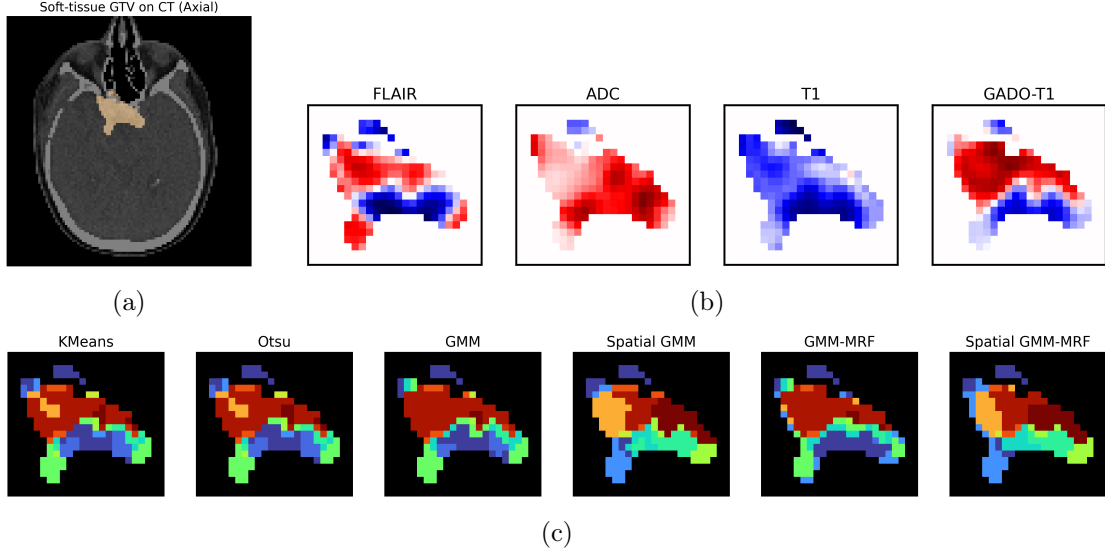


Figure 5: a) Soft tissue GTV mask. b) Normalized image intensities within the soft tissue GTV for four MRI sequences. c) Habitats formed by clustering all images into high- and low- regions and labeling each voxel with its high/low combination across sequences. Six different clustering approaches were used.

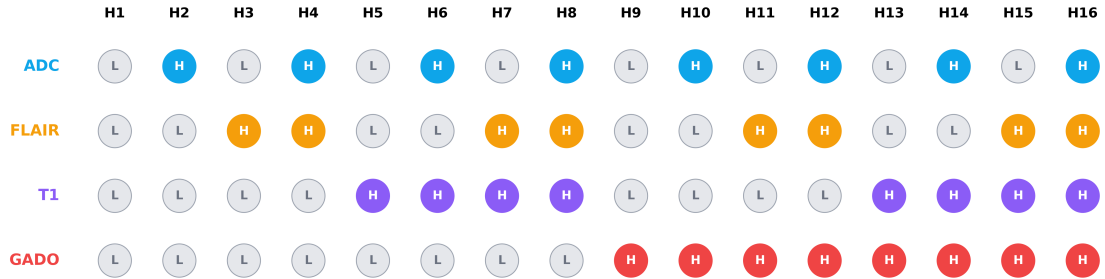


Figure 6: Binary encoding of habitat labels using four MRI sequences. Circles labelled “H” indicate the voxel belongs to the high-intensity cluster for the corresponding sequence, while “L” indicates the low-intensity cluster.

2.3 Feature Extraction

Following habitat construction, features were extracted for each clustering method and patient. For every habitat, the absolute volume within the soft-tissue GTV was calculated. Relative habitat volume fractions were then computed by dividing the habitat volume by the total soft-tissue GTV volume, providing a normalized representation of the spatial distribution of habitats within the soft tissue GTV. Volume fractions were

preferred to absolute volumes, since they captured the GTV distributions without introducing large inter-patient differences.

For each habitat and clustering method, statistical measures including the mean, median, standard deviation, and interquartile range (IQR) were calculated. To assess the differences between the patients who experienced local failure (LF group) or not (no LF group), a Mann–Whitney U test was performed for each habitat. To account for multiple comparisons and the false discovery rate, p-values were corrected using the Benjamini-Hochberg procedure, and significance level of 0.05 was set.

In addition to habitat-derived features, relevant clinical variables were collected for each patient. These included the patient age at the time of treatment and the time-to-event, defined as either the time to local failure for patients experiencing recurrence or the time to last clinical follow-up for patients without recurrence. Sex was discarded as a clear imbalance in the population was observed between the LF vs. no LF groups.

2.4 Survival Analysis

The extracted habitat volume fractions were combined with clinical data to construct a survival analysis dataset of covariates, with local failure as the outcome of interest. An additional dataset of covariates was created, where volume fractions were replaced by a binary representation of habitat presence. The statistical procedure detailed in the following sections was applied to both dataset, yielding two CPH models. Implementation was carried out using the scikit-survival Python library [31].

Data preparation

Prior to performing the survival modelling, habitat volume fractions were preprocessed. Because the fractions are compositional information of the GTV, each patient’s volume fractions form a compositional vector $x = (x_1, \dots, x_D)$ of non-negative values that sum to one, where D denotes the number of habitats and x_i represents the fraction of the i -th habitat. This introduces a dependency between features, as an increase in one habitat implies a decrease in at least one other habitat. As a result, the raw volume fractions violate the independence assumptions required by the CPH model. To address this issue, a compositional data transformation based on the centered log-ratio (CLR) [32],

$$\text{clr}(x) = \ln \left[\frac{x_1}{g_m(x)}, \dots, \frac{x_D}{g_m(x)} \right], \quad (10)$$

was applied using the scikit-bio Python library [33]. The CLR transformation expresses each component as the logarithm of its ratio to the geometric mean of all components,

$$g_m(x) = \left(\prod_{i=1}^D x_i \right)^{1/D}. \quad (11)$$

This transformation maps the compositional data into an unconstrained log-ratio space, mitigating the dependencies introduced by the constant-sum constraint. This results

in covariates that better satisfy the assumptions of the CPH model while preserving the relative information between habitats. Before the transformation, zero values were replaced using the multiplicative replacement strategy, which replaces zero values with a small positive constant δ while proportionally shrinking the non-zero parts to preserve a total sum of one, avoiding undefined logarithmic operations [34].

For the second CPH model, the prevalence of habitats was defined according to a minimum volume criterion. A volume fraction below of 5% was considered negligible and prevalence was set to 0. Similarly, habitats appearing in less than 10% of patients were discarded.

Cox proportional hazards (CPH) model

After preprocessing, the habitat features (volume fractions or habitat prevalence variables) were analyzed using a penalized survival model to identify the most informative predictors of local failure. Since the number of covariates was relatively high compared to the cohort size and number of observed events, regularized regression was required to reduce overfitting and improve model interpretability. For this purpose, an L1-regularized CPH model was employed.

To perform habitat selection and improve model robustness, L1 regularization was introduced into the Cox regression framework by using the least absolute shrinkage and selection operator (LASSO) penalty. The penalized optimization problem is defined as

$$\mathcal{L}_{\text{penalized}} = -\ell(\beta) + \alpha \sum_{j=1}^p |\beta_j| \quad (12)$$

where $\ell(\beta)$ is the Cox partial log-likelihood, β_j are the regression coefficients, p is the number of habitats, and α controls the strength of the penalty. The L1 penalty shrinks non-informative coefficients to zero, thereby retaining only the most relevant habitats in the final model. Larger values of α impose stronger regularization and produce models with fewer selected features, whereas smaller values allow more covariates to remain in the model.

To determine the optimal regularization strength, repeated and stratified cross-validation was performed, preserving the proportion of local failure cases across folds. For each candidate α among a range of candidate values, predictive performance was evaluated using the concordance index (C-index), which quantifies the agreement between predicted risk and observed survival outcomes. The optimal α was selected as the value maximizing the mean cross-validated concordance index. Habitats with non-zero hazard ratios at this optimal penalty level were retained as initially selected candidates.

To assess the robustness of habitat selection, a bootstrap stability analysis with 2000 iterations was subsequently performed using the previously selected optimal α . In each bootstrap iteration, the dataset was resampled in a stratified manner with respect to

event status to preserve the proportion of observed local failures, with each bootstrap sample containing 5 local failure cases and 24 non-local failure cases (80% of each subgroup). The L1-penalized CPH model was then fitted, and the selected features were recorded. Habitat stability was measured as the proportion of bootstrap iterations in which a habitat obtained a non-zero coefficient, providing a measure of robustness with respect to sampling variability.

Habitat stability was measured as the proportion of bootstrap iterations in which a habitat obtained a non-zero coefficient, providing a measure of robustness with respect to sampling variability. In addition to selection frequency, the distribution of Cox coefficients obtained across bootstrap iterations was analyzed to evaluate directional consistency of habitat effects (Figs. 8 and 9). For each habitat, the 25th and 75th percentiles of the coefficient distribution were computed. A sign-consistency criterion was then applied, requiring both quartiles to share the same sign, ensuring that the habitat consistently behaved as either risk-increasing or risk-decreasing across bootstrap resamples. Habitats satisfying both a stability threshold of at least 50% selection frequency and the sign-consistency criterion were defined as robust habitats.

Finally, the set of robust habitats was used to obtain an L1-penalized CPH model on the full dataset. Cross-validation over the regularization path was repeated to obtain the final optimal α . The resulting model yielded the final set of selected habitats and their associated hazard coefficients, as well as the final C-Index. This complete procedure was independently repeated for each clustering technique for two different CPH models:

- CPH model using habitat volume fractions as covariates.
- CPH model using binary habitat presence as covariates.

Final feature selection

To obtain an overall measure of habitat significance across clustering strategies, the results from all six clustering method-specific survival models were aggregated into a single score for each habitat. For every clustering method, the selected habitat coefficients, $\beta_{h,m}$, their bootstrap-derived stability values, $s_{h,m}$, and the predictive performance of the corresponding model, quantified by the concordance index, C_m , were retrieved. A final “importance” score, I_h , was then computed for each habitat by summing the contribution of that habitat across all clustering methods:

$$I_h = \sum_{m=1}^M (\beta_{h,m} \cdot s_{h,m} \cdot C_m). \quad (13)$$

Through this formulation, habitats consistently selected across multiple clustering approaches, exhibiting stable coefficient behaviour, and originating from models with stronger predictive performance obtained larger scores. The resulting score therefore reflects the overall relevance and robustness of each habitat with respect to local failure prediction

across all clustering-derived habitat representations. The same formula was used for both CPH model's, resulting in two scores per habitat, one for volume fraction and one for presence.

Habitat co-selection

Using the results of the bootstrap stability analysis, relationships between habitats were further investigated by analysing their patterns of joint selection across bootstrap iterations. For each iteration, if the C-index was above 0.9, the frequency with which pairs of initially selected habitats were selected together was quantified as a percentage of co-selection. These values were then arranged into a co-selection matrix, where each entry represents the proportion of bootstraps in which two habitats were simultaneously selected. A graph representation of habitat interactions was then constructed, in which nodes correspond to habitats and edge thickness is proportional to their co-selection frequency. All edges are shown, but only co-selection frequencies above 40% are labelled at the mid-point between connecting nodes. Graphs for all six clustering techniques were created for the volume fraction and habitat presence CPH models. This representation enabled further exploration of potential dependencies and structural relationships between habitats beyond their individual selection stability.

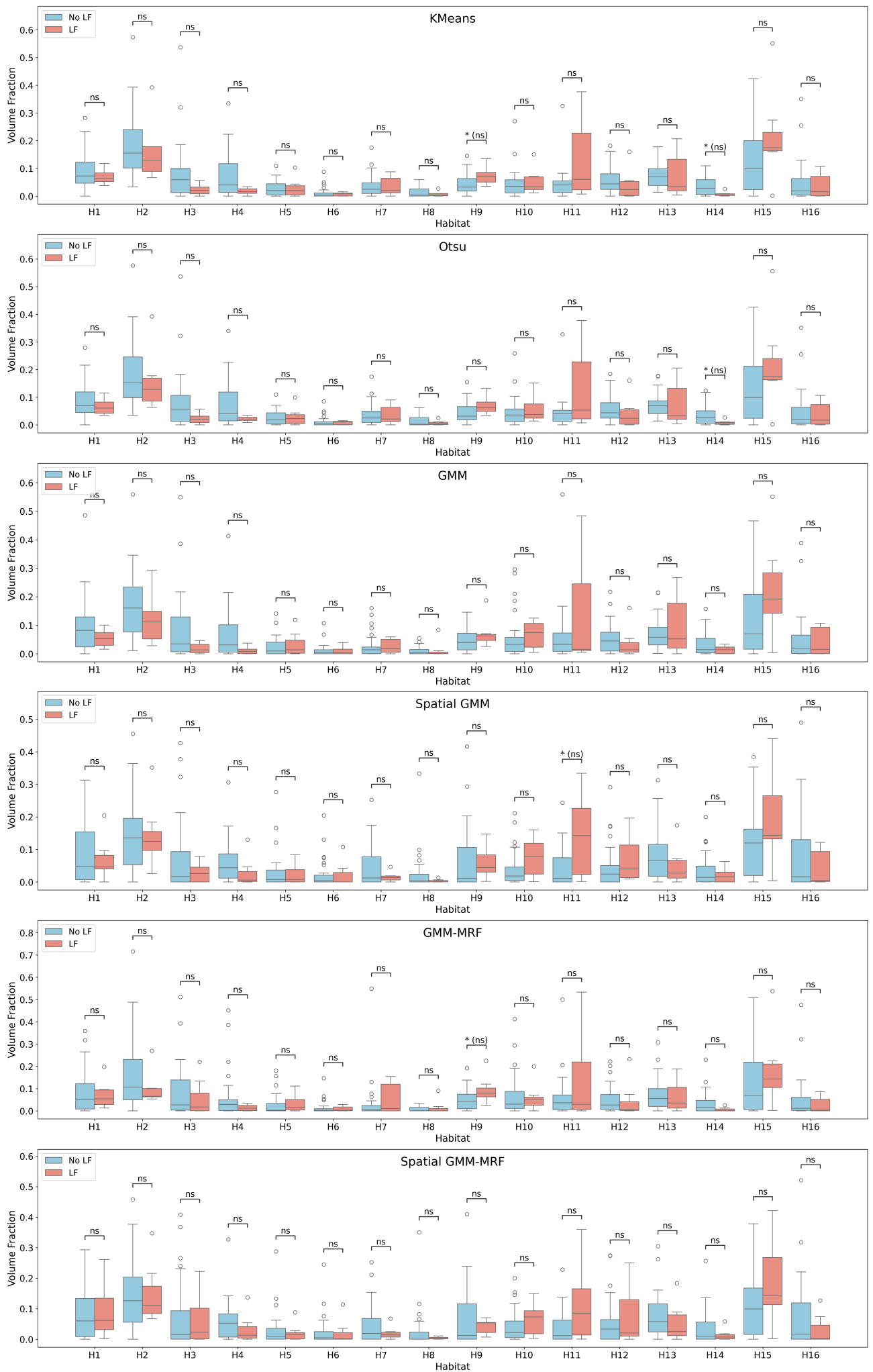


Figure 7: Volume fraction distributions between local failure (LF) and no LF patients across clustering approaches. Mann–Whitney U-test was performed to identify significance (0.05 significance threshold).

3 Results

3.1 GTV Habitat Distributions

The habitat volume fraction distributions for LF and no LF patient groups for all clustering methods are shown in Fig 7. LF and No LF volume fractions within each habitat were compared using the Mann Whitney U-tests. Since volume fractions were non-normal, bounded at zero, and potentially contained ties, p-values were computed using a tie-corrected asymptotic approach, and were adjusted for multiple habitat-wise comparisons. This statistical test looks for differences in distributions or stochastic dominance, and not whether the means differ. Although some visual differences in the distributions are apparent, none of the habitat volume fractions demonstrated statistically significant differences in distributions between the two patient groups. This may be attributed to the limited sample size and the influence of outliers within individual groups. Detailed summary statistics, including the mean, median, standard deviation, and IQR are provided in Appendix A.

3.2 Cox Proportional Hazards Model Results

The distributions of Cox coefficients resulting from the robustness analysis using bootstrapping for the model using volume fractions and the model using habitat presence can be seen in Figs. 8 and 9, respectively. The boxplots show the median, IQR and standard deviation of the coefficient’s distribution for each habitat. High variability of both the selected habitats and the number of selected habitats was observed between different clustering methods, while most habitats showed consistent directionality.

Additionally, Tables 1 and 2 show the selected habitats and their respective regression coefficients for the CPH models using volume fractions and habitat presence, respectively. Negative coefficients indicate hazard-decreasing habitats, while positive coefficients represent an increased hazard of local failure. Habitats 4 (high ADC and high FLAIR) and 9 (high GADO-T1) were the most consistently selected habitats across both models, with negative and positive weight, respectively. Their consistency in coefficient sign is important, as biologically meaningful and robust features are expected to maintain the same directional effect across clustering methods and models, either acting consistently as hazard-decreasing or as risk-increasing habitats. In the model using volume fractions, habitats 3 (high FLAIR) and 12 (low T1) were also selected by three clustering methods and consistently showed risk-decreasing effects, identifying them as additional potential features of interest. In contrast, although habitat 7 (high FLAIR and high T1) was also selected by three clustering methods in the habitat presence model, the change of sign in the regression indicates an unstable association with outcome, and therefore questionable reliability.

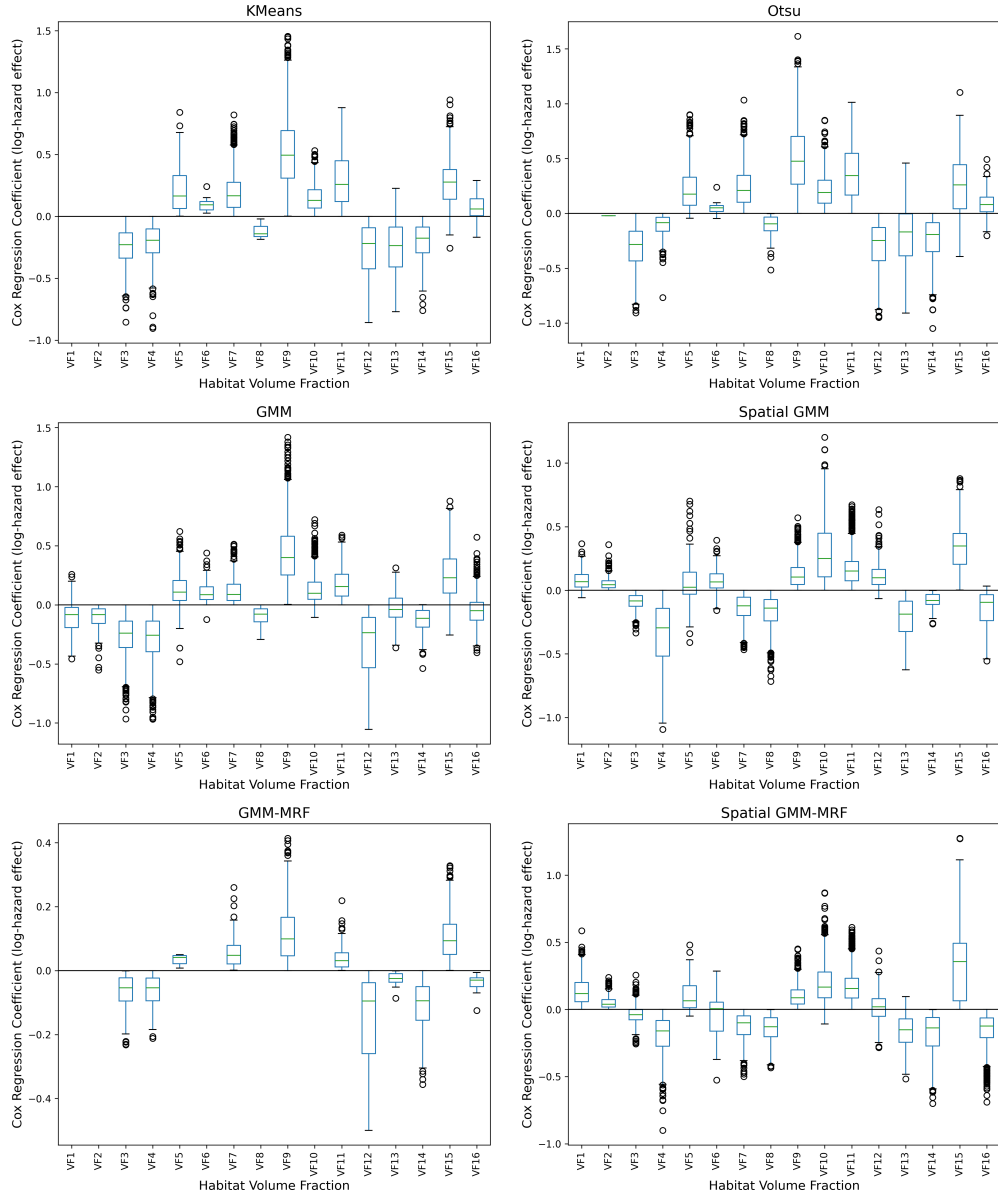


Figure 8: Distribution of Cox coefficients using volume fractions as model covariates.

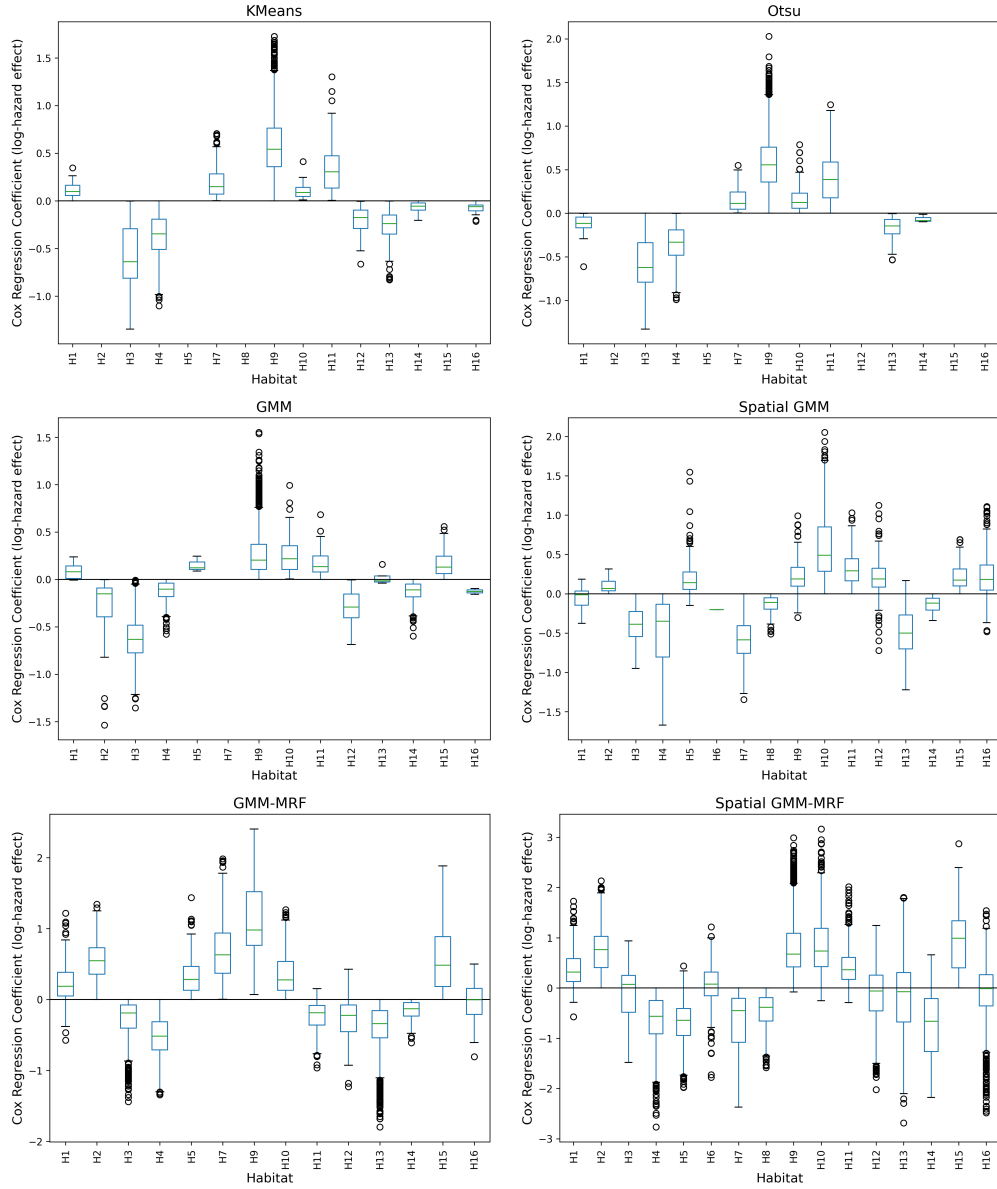


Figure 9: Distribution of Cox coefficients using habitat presence as model covariates.

Model performance

The predictive performances of the final Cox proportional hazards models, evaluated using the concordance index (C-index), are shown in Fig. 10. For the volume fraction models, KMeans clustering achieved the highest predictive performance, followed by GMM-MRF and Otsu thresholding. As for the habitat presence models, KMeans and Otsu obtained the highest C-indices, while GMM-MRF again showed competitive per-

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12	H13	H14	H15	H16
KMeans			-0.20	-0.05					0.60			-0.05				
Otsu			-0.28						0.41	0.05	0.12	-0.21		-0.06		
GMM			-0.22	-0.15					0.45			-0.11				
GMM-MRF									0.08							
Spatial GMM				-0.20				-0.05		0.09						
Spatial GMM-MRF				-0.07				-0.14								

Table 1: Final Cox regression coefficients across clustering methods using volume fractions as covariates. Only selected habitats are shown.

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12	H13	H14	H15	H16
KMeans				-0.97					0.58							
Otsu				-0.96					0.56							
GMM			-0.34													
GMM-MRF		0.61		-0.59			0.41		1.16						0.09	
Spatial GMM				-0.28			-0.91			0.29						
Spatial GMM-MRF		0.74		-0.31			-0.28		0.35	0.45						

Table 2: Final Cox regression coefficients across clustering methods using habitat presence as covariates. Only selected habitats are shown.

formance. In contrast, the spatial GMM and spatial GMM-MRF approaches reportedly scored low in both modelling strategies.

Overall, clustering methods based on simpler intensity partitioning approaches, such as KMeans and Otsu thresholding, demonstrated more robust predictive performance than the spatially constrained alternatives. The incorporation of MRF-based spatial regularization improved the performance of standard GMM clustering. Additionally, habitat presence models generally outperformed by a small margin the corresponding volume fraction models across clustering methods. These results indicate that the choice of clustering strategy may influence the prognostic value of the derived habitats to an extent that is clinically relevant.

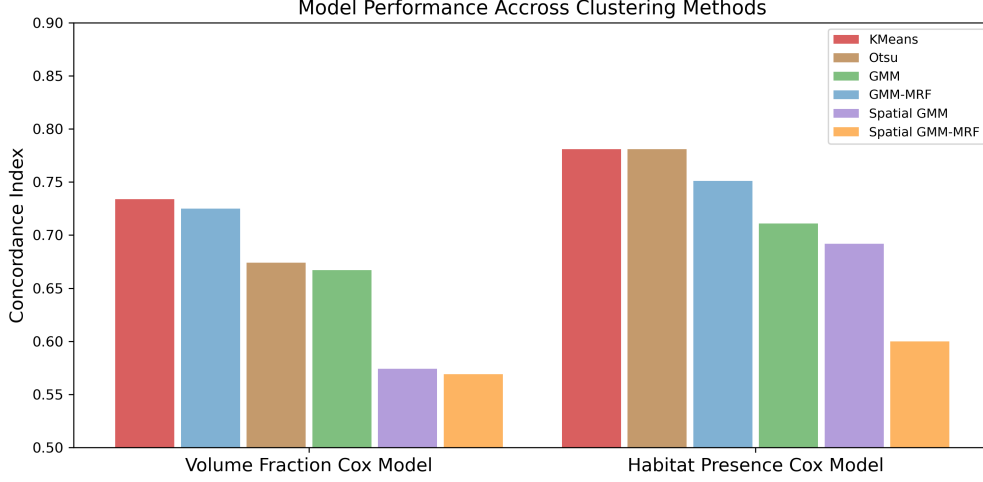


Figure 10: Concordance indices of all clustering methods for Cox proportional hazards models using volume fractions and habitat presence as covariates.

Key habitats

Habitat 9 (high GADO-T1) shows the highest hazard-increasing “importance” score, as defined in Eq. 13, in both models using volume fractions and habitat presence, as shown in Figs. 12 and 11, respectively. In contrast, habitats 3 (high FLAIR) and 4 (high FLAIR and high ADC) consistently showed negative “importance” scores, indicating a hazard-decreasing effect. These habitats demonstrated concordant coefficient directions across multiple clustering approaches and across both feature representations, suggesting an underlying true signal not dependent on a specific clustering strategy or model. Additionally, habitat 10 (high ADC and high GADO-T1) also exhibited consistent coefficient directionality across models, although its “importance” scores remained comparatively low, indicating a weaker contribution to prediction performance.

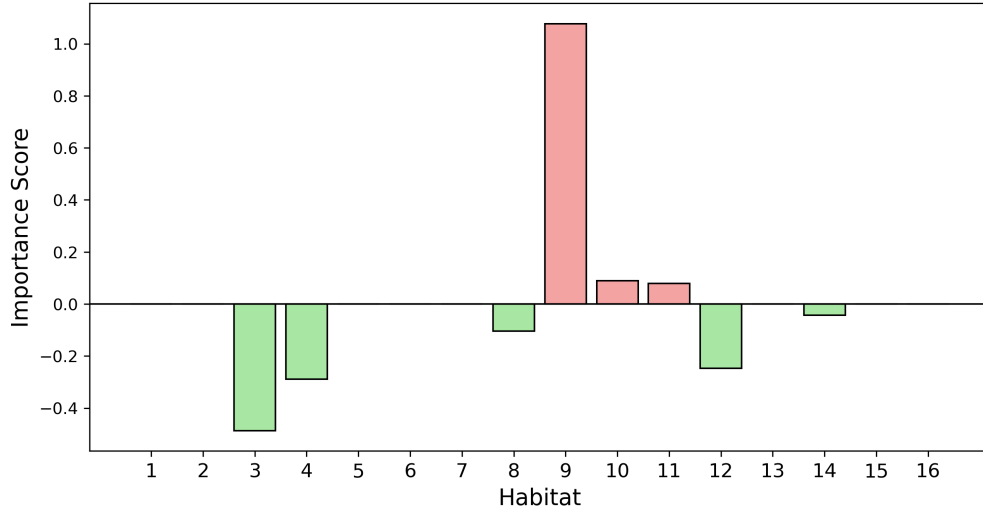


Figure 11: Habitat scores obtained applying Eq. 13 for the Cox proportional hazards model using volume fractions.

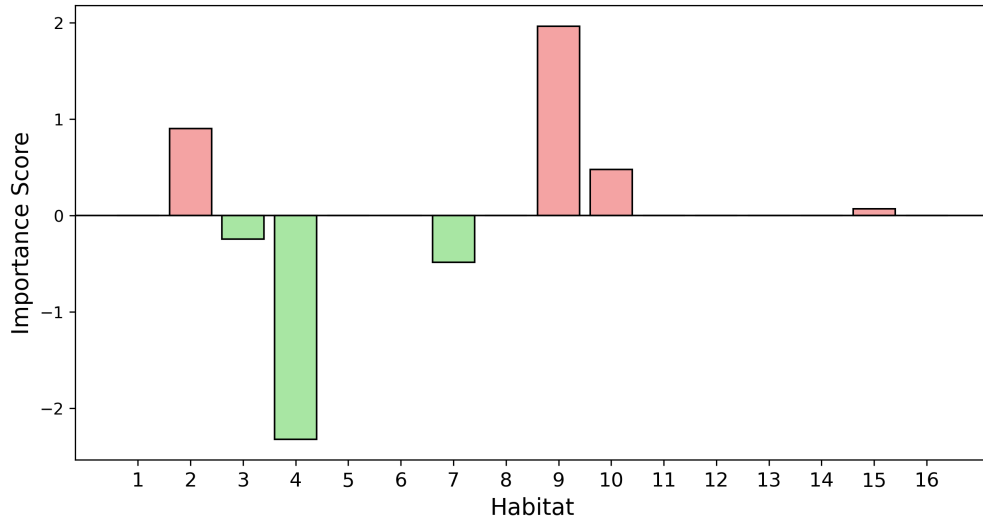


Figure 12: Habitat scores obtained applying Eq. 13 for the Cox proportional hazards model using habitat presence.

Habitat co-selection

The frequency of habitats co-occurring across bootstrap was investigated for runs with C-indices above 0.9 in order to identify potential pairs of habitats that may act together as a single feature. Since KMeans clustering yielded the highest C-index among the evaluated clustering methods (Figs. 11 and 12), this approach is put forward as an exemplary case.

Corresponding co-selection graphs for both CPH models are shown in Fig. 13. Only habitats co-selected in more than 40% of the bootstraps are shown. Habitats 4,9, and 3 emerge as frequently selected together, showing the importance of not only considering habitats individually, but as a connected network that takes into account correlations and dependencies across the habitats. The co-selection graphs for all clustering methods are provided in the App. B.

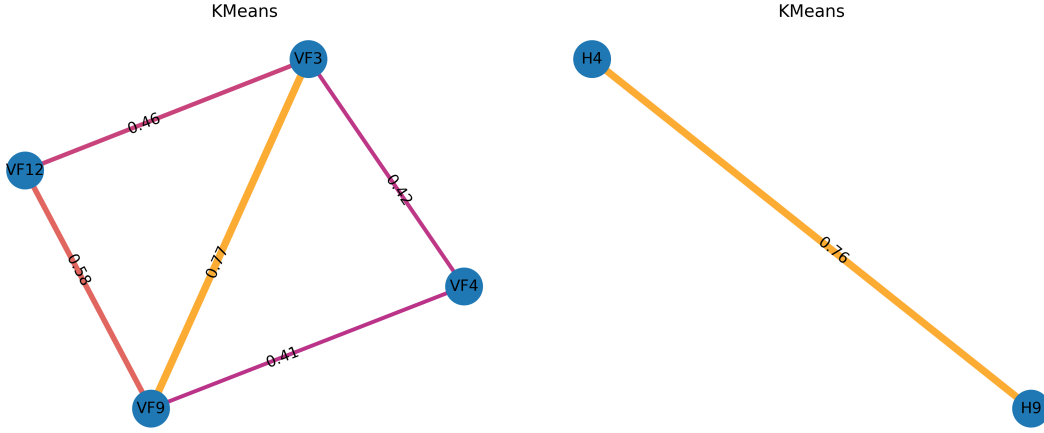


Figure 13: Habitat Co-selection network using K-Means clustering for both the volume fraction (left) and presence (right) CPH models. Only habitats with a co-selection frequency of more than 40% of the bootstrap iterations with C-indices above 0.9 are shown.

4 Discussion

4.1 Interpretation of Results

GTV volume fractions

Differences in the habitat volume fraction distributions (Fig. 7) between patients experiencing local failure and those without recurrence were not statistically significant, when correcting p-values for multiple comparisons. The absence of significant results is likely strongly influenced by the limited cohort size, particularly the small number of local failure cases ($n = 7$, ca. 12% of the cohort). Statistical comparisons between groups becomes challenging when one population is represented by only a few samples, as the resulting distributions are highly sensitive to individual patient variability and outliers. In this context, the Mann-Whitney U test may not be able to detect small but potentially relevant differences in habitat compositions. Furthermore, the large number of tested habitats relative to the number of observed events increases the impact of multiple-comparison correction procedures, further reducing the likelihood of significance.

Impact of clustering techniques on the concordance index

Despite the lack of statistically significant differences in individual habitat distributions between responding and local failure patients, the survival analysis demonstrated that habitat-derived features contain predictive information when used in combination as a habitat signature. Among the evaluated clustering approaches, KMeans achieved high concordance indices for both the volume-fraction and habitat-presence regression models. Otsu thresholding and GMM-MRF also showed competitive predictive performance, whereas spatial GMM and spatial GMM-MRF performed worse.

The high concordance index obtained using KMeans and Otsu clustering suggests that relatively simple intensity partitioning strategies can capture clinically relevant intra-tumour heterogeneity in this cohort. Both methods segment the tumour into high- and low-intensity subregions, potentially preserving discriminative intensity patterns without introducing excessive model complexity. Similarly, the improved performance of GMM-MRF compared to standard GMM indicates that moderate spatial regularization may help reduce isolated noisy voxels and produce more spatially coherent habitat structures while still preserving meaningful local heterogeneity.

In contrast, the lower performance of spatially aware GMM approaches may indicate that stronger neighbouring constraints reduce sensitivity to the imaging biomarker signals required for habitat analysis. By explicitly incorporating voxel coordinates during clustering, these methods favour smoother and more spatially continuous regions, which however may suppress small but relevant tissue-intensity variations. Assuming that the observed prognostic value of habitat analysis reflects a true biological signal rather than a coincidental finding, these results suggest that, in the present dataset, preserving local intensity heterogeneity may be more important for outcome prediction than enforcing strong spatial continuity, as excessive spatial smoothing may dampen the predictive signal.

Across all clustering strategies, the CPH models using habitat-presence outperformed the habitat-fraction ones. This suggests that the mere presence of certain habitats may be more informative for local failure prediction than their relative volumetric contribution, or may reflect the difficulties in obtaining reliable habitat volume regions. Additionally, volume fractions introduce further compositional complexity, as increases in one habitat necessarily imply decreases in others. Although centered log-ratio transformation (Eq. 10) was applied to mitigate this issue, the resulting variables may remain sensitive to noise and small volumetric fluctuations. In contrast, binarizing habitat presence simplifies the representation and may improve robustness in small datasets by reducing the influence of outliers and measurement variability.

Habitat selection

Habitats 4 and 9 emerged as the most consistently selected predictors across clustering methods and survival CPH models. Additionally, habitats 3 and 10 demonstrated lower, but still consistent, predictive relevance. The following interpretations of the identified

habitats are exploratory, and due to the complexity of multi-parametric MRI signal interpretations, supported by AI-assisted synthesis.

Habitat 9 represents regions with high GADO-T1 intensity alone, corresponding to contrast-enhancing tumour areas with increased vascular permeability and disruption of the blood-brain barrier. Such regions are frequently associated with active tumour tissue, angiogenesis, and aggressive disease behaviour. The consistent positive hazard coefficients observed for habitat 9 across multiple clustering methods therefore align with established radiological interpretations of gadolinium enhancement as a primary source for tumour delineation and potentially a marker of tumour aggressiveness.

Habitat 4, characterized by high ADC and high FLAIR signal, was repeatedly identified as a risk-decreasing feature, particularly in the habitat-presence model. Elevated ADC values are generally associated with lower cellular density, necrosis, or increased extra-cellular space, while high FLAIR intensity often reflects oedema, infiltrative changes, or treatment-related tissue alterations. The combination of these imaging characteristics may therefore correspond to less densely cellular tumour regions rather than highly aggressive viable tumour. The consistent negative hazard coefficients observed suggest that the presence or greater representation of this habitat may be associated with a lower risk of local failure. However, FLAIR hyperintensity may also arise at the surgical site as a consequence of intervention; therefore, this imaging feature may not necessarily reflect an intrinsic tumour characteristic, but could instead act as a proxy for surgical complexity and associated post-interventional changes.

Habitat 3, defined primarily by high FLAIR intensity, also showed a stable hazard decreasing effect across clustering approaches. High FLAIR signal may reflect infiltrative tumour margins, oedema, or non-enhancing abnormal tissue surrounding the enhancing core, but may also arise from post-surgical or treatment-related changes. Therefore, this habitat may not necessarily represent an intrinsic tumour phenotype. Although FLAIR abnormalities are not necessarily benign, the absence of strong enhancement or aggressive diffusion characteristics may indicate tissue states with lower immediate treatment resistance compared to actively enhancing tumour regions. Alternatively, the repeated selection of habitat 3 with negative coefficients may suggest that this feature captures information related to surgical or treatment complexity rather than tumour biology alone.

Habitat 10, characterized by high ADC and high GADO-T1 signal, showed a consistent but weaker hazard-increasing effect. This habitat may represent mixed tissue states containing both enhancing tumour components and regions of reduced cellular density or partial necrosis. While its contribution to the predictive models was less prominent, the repeated consistency in coefficient direction suggests that it may still reflect biologically relevant tumour characteristics associated with local failure risk.

Several habitats were not selected by the survival models, including habitats 5, 6, 8, 11, 12, 13, 14, and 16 in the habitat presence analysis, and habitats 1, 2, 5, 6, 7, 13, 15, and 16 in the volume fraction model. This may indicate that these combinations of imaging characteristics either occur infrequently within the cohort or do not contribute substan-

tially to local failure prediction. Additionally, some habitats may represent biologically neutral tissue states lacking strong association with treatment resistance. Such habitats may be excluded from future studies to reduce the number of covariates.

Co-selection of habitats

The co-selection analysis revealed distinct patterns of habitat interactions across the different clustering methods and modelling approaches. For the K-means clustering method, which achieved the highest predictive performance, the volume fraction model demonstrated a relatively complex co-selection structure involving multiple interconnected habitats. In particular, habitats 4, 9, 3, and 12 showed the strongest and most consistent co-selection frequencies. Based on the habitat definitions, these habitats are characterized by elevated FLAIR and/or GADO-T1 signal intensities, with habitats 4 and 12 additionally presenting high ADC signal. The repeated co-selection of these habitats may therefore suggest that regions associated with oedema-like abnormalities, contrast enhancement, and altered diffusion properties jointly contribute to local failure risk, potentially reflecting an underlying microenvironment captured across complementary imaging modalities.

For the habitat presence model, a particularly strong co-selection relationship was observed between habitats 9 and 4. While habitat 9 is characterized by isolated high GADO-T1 signal, habitat 4 combines high ADC and high FLAIR signal. Their frequent joint selection may therefore suggest that the coexistence of contrast enhancing tumour regions with regions characterized by FLAIR-related abnormalities and altered diffusion properties contains prognostic information. However, given the potential contribution of post-surgical or treatment-related FLAIR changes, this association should be interpreted cautiously and may not reflect tumour biology alone.

Across all clustering methods (App. B), the co-selection graphs demonstrated substantial variability in both network complexity and dominant habitat pairs. Some methods, such as Spatial GMM-MRF, produced dense interaction networks with numerous moderate co-selection relationships, whereas others, including standard GMM-MRF for the volume fraction model and GMM for the presence model, yielded structures with only one habitat. Nevertheless, recurrent involvement of habitats characterized by elevated FLAIR and/or GADO-T1 signal intensities was observed across several methods, further suggesting that imaging features related to oedema-like abnormalities and contrast enhancement consistently contribute to the predictive models irrespective of the clustering method used. Overall, these findings indicate that co-selection analysis may provide additional insight into relationships between habitats and the potential cooperative role of multiple tumour phenotypes in outcome prediction.

4.2 Limitations

The primary limitation of this study was the relatively small cohort size, particularly the low number of observed local failure events relative to control cases. Survival models are

highly sensitive to the ratio between the number of covariates and the number of events, and limited event counts increase the risk of overfitting and unstable feature selection. Although L1-regularization and bootstrap-based stability analysis were used to reduce these effects, larger cohorts would be necessary to validate the robustness and validity of the identified habitats.

Another important limitation of this study was the absence of dosimetric information in the survival analysis. Since local failure after radiotherapy is strongly influenced by the delivered dose distribution, some of the predictive signal attributed to imaging-derived habitats may be coincidental, and actually reflect clinical underdosage due to planning constraints. Additionally, no dosimetry-only baseline model was included, making it difficult to assess whether the habitat features provide complementary prognostic value beyond standard dose-related predictors. Future studies could therefore integrate dosimetric variables and compare combined imaging-dosimetry models against baseline models.

Lastly, the study was partially restricted by the use of two-class, high/low clustering for each MRI sequence. While binary separation simplifies habitat interpretation and reduces the model dimensions, it may oversimplify complex tumour biology and fail to capture subtle imaging phenotypes.

4.3 Uncertainty

Several sources of uncertainty may influence the robustness and reproducibility of the proposed habitat framework. Firstly, and probably the most important source of uncertainty, is image registration. Since all MRI sequences were aligned to the planning CT, inaccuracies in registration may lead to spatial mismatches between modalities and consequently alter habitat definitions at the voxel level. This is particularly relevant for DWI/ADC images, which required an additional intermediate registration step.

Another major source of uncertainty arises from GTV contouring. Tumour delineation was performed manually and for the purpose of treatment planning with above-millimetre safety margins. Despite the clinical experience of the oncologists, it is subject to inter-observer variability, and uncertainty at the tumour boundaries. Since habitats are generated exclusively within the GTV, small contour differences may substantially affect their composition, especially for small or fragmented subregions, or tumours that strongly overlap with additional body structures like bone regions.

Clustering uncertainty also contributes to variability in habitat generation. Different clustering algorithms may produce substantially different spatial representations of the same tumour. Even under the same clustering framework and settings, initialization sensitivity, probabilistic assignment, and smoothing constraints may influence the final habitat configuration. The lack of a biological ground truth for habitat definition further complicates the interpretation of clustering-derived regions.

Additional uncertainty may arise from MRI acquisition variability, including acquisition protocols and image noise. Although robust normalization procedures were applied, MRI

intensities remain inherently nonstandardised.

4.4 Future Work

Several directions for future research emerge from this work. One promising avenue involves further analysis of habitat interactions and synergistic contribution. Habitat co-selections of 3 or more habitats may provide insight into whether combinations of habitats collectively describe meaningful tumour ecosystems rather than acting as isolated predictors.

An additional direction for future work could explore the spatial distribution of habitats within the tumour. In this study, habitats were treated primarily as global volumetric fractions, without explicitly considering their spatial organisation. However, the spatial arrangement between habitats may carry additional relevant information, as neighbouring or interacting regions could reflect underlying tumour processes, such as diffusion gradients or invasive growth patterns. Incorporating spatial relationships between habitats could therefore provide a more detailed characterisation of intra-tumour heterogeneity and potentially improve outcome modelling.

Future work could also expand the habitat framework through the inclusion of additional imaging modalities or MRI sequences. In particular, T2-weighted imaging may provide complementary information regarding tissue composition that is not fully captured by FLAIR or T1 alone. Other advanced MRI techniques, such as perfusion-weighted imaging (PWI), could further improve the characterization of tumour heterogeneity. Additionally, future studies could involve the integration of longitudinal imaging. Rather than analysing habitats only at baseline, temporal habitat evolution during or after treatment could provide valuable information regarding treatment response dynamics and emerging resistant subregions.

Finally, external validation on larger and independent cohorts would be essential to assess the reproducibility and clinical utility of the identified habitats. Combining habitat-based imaging features with histopathological information may additionally enable the development of more comprehensive multi-modal predictive models for personalized proton therapy outcome modelling.

5 Conclusion

This work presented a multi-modal MRI habitat generation pipeline for the analysis and prediction of local failure following proton therapy. By combining information from ADC, FLAIR, T1, and GADO-T1 MRI sequences, voxel-wise habitats representing distinct imaging phenotypes were generated using multiple clustering strategies. The resulting habitat-derived features, including habitat volume fractions and habitat presence, were incorporated into survival analysis CPH models to evaluate their prognostic potential. Although no statistically significant differences in the volume fractions of individual habitats were observed between local failure and nonlocal failure groups, the survival

modelling demonstrated that specific habitats consistently contributed to outcome prediction despite differences in clustering approaches.

Among the evaluated methods, intensity-based clustering approaches achieved the strongest predictive performance, suggesting that spatial blurring does not substantially improve the habitat representations of tumour heterogeneity. In particular, habitats associated with high FLAIR and T1 signal, as well as regions of high GADO-T1, emerged as the most consistent predictors of local failure. Despite limitations related to cohort size and imaging variability, the proposed framework demonstrates the potential of habitat imaging as a tool for capturing relevant intra-tumour heterogeneity features. Future work involving larger cohorts, additional imaging modalities, habitat synergies, and longitudinal analysis may further improve the robustness and clinical applicability of habitat-based outcome modelling in proton therapy.

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Appendix

A Habitat Volume Fraction Statistics

Habitat	LF				No LF			
	Mean (%)	SD (%)	Median (%)	IQR (%)	Mean (%)	SD (%)	Median (%)	IQR (%)
1	7	3	6	3	9	6	7	8
2	16	11	13	9	19	12	16	14
3	2	2	2	2	8	11	6	9
4	2	1	2	2	7	8	4	10
5	3	4	2	3	3	3	2	4
6	1	1	1	1	1	2	0	1
7	4	3	2	5	4	4	2	4
8	1	1	0	1	2	2	0	3
9	7	3	7	3	4	4	3	5
10	5	5	3	5	5	5	4	5
11	14	15	6	20	5	6	4	4
12	4	6	2	5	6	5	4	6
13	8	8	3	11	7	4	7	6
14	1	1	1	1	4	4	3	5
15	22	17	18	7	13	13	10	18
16	4	5	2	7	5	8	2	6

Table A1: Habitat volume fraction statistics using KMeans clustering.

Habitat	LF				No LF			
	Mean (%)	SD (%)	Median (%)	IQR (%)	Mean (%)	SD (%)	Median (%)	IQR (%)
1	7	3	6	4	9	6	7	7
2	16	11	13	8	19	12	15	15
3	2	2	2	2	8	11	6	9
4	2	1	2	1	7	8	4	10
5	3	3	2	3	3	3	2	4
6	1	1	1	1	1	2	0	1
7	4	3	2	5	4	4	3	4
8	1	1	0	1	2	2	0	3
9	7	3	6	3	4	4	3	5
10	6	5	4	5	5	5	4	5
11	13	15	5	21	5	6	4	4
12	4	6	2	5	6	5	4	6
13	8	8	3	11	7	4	7	5
14	1	1	1	1	4	4	3	4
15	22	17	18	8	14	13	10	19
16	4	5	2	7	5	8	2	6

Table A2: Habitat volume fraction statistics using Otsu thresholding.

Habitat	LF				No LF			
	Mean (%)	SD (%)	Median (%)	IQR (%)	Mean (%)	SD (%)	Median (%)	IQR (%)
1	5	3	5	4	10	10	8	11
2	12	9	11	10	17	12	16	16
3	2	2	1	3	9	12	4	12
4	1	1	1	2	7	9	3	10
5	3	4	1	5	3	3	1	4
6	1	1	0	2	1	2	0	1
7	3	3	2	5	3	4	1	2
8	1	3	0	1	1	1	0	2
9	7	5	6	2	5	4	4	6
10	7	5	7	8	6	8	3	5
11	15	19	2	23	6	10	3	6
12	4	6	1	3	5	5	4	7
13	10	12	5	16	7	5	6	6
14	1	1	1	2	4	4	1	5
15	23	17	19	14	13	14	7	19
16	4	5	2	9	5	9	2	6

Table A3: Habitat volume fraction statistics using GMM clustering.

Habitat	LF				No LF			
	Mean (%)	SD (%)	Median (%)	IQR (%)	Mean (%)	SD (%)	Median (%)	IQR (%)
1	7	7	5	4	8	9	5	15
2	14	10	13	6	15	12	14	14
3	3	3	3	5	7	12	2	9
4	3	5	1	3	6	7	4	7
5	2	3	1	4	3	6	1	4
6	2	4	0	3	2	5	0	2
7	2	2	1	1	5	6	1	8
8	0	0	0	0	3	6	0	2
9	6	5	4	5	6	10	1	11
10	8	6	8	9	4	5	2	4
11	14	13	14	20	4	6	1	7
12	7	7	4	10	5	7	2	5
13	5	6	3	5	8	8	7	10
14	2	2	2	3	4	5	1	5
15	20	15	14	13	12	11	12	14
16	5	6	0	9	8	12	2	13

Table A4: Habitat volume fraction statistics using Spatial GMM clustering.

Habitat	LF				No LF			
	Mean (%)	SD (%)	Median (%)	IQR (%)	Mean (%)	SD (%)	Median (%)	IQR (%)
1	7	6	5	7	9	10	6	11
2	10	8	7	4	16	16	11	18
3	6	9	2	8	9	12	3	14
4	1	1	1	2	6	11	3	5
5	3	4	2	5	2	4	0	3
6	1	1	0	2	1	3	0	1
7	6	7	1	12	2	3	0	2
8	2	3	0	1	1	1	0	1
9	9	6	8	4	5	5	4	7
10	6	7	5	4	7	10	3	7
11	14	20	3	21	6	10	4	7
12	5	9	1	4	5	6	3	7
13	7	8	4	9	7	7	5	8
14	1	1	0	1	4	5	2	5
15	19	17	14	11	13	16	5	22
16	3	4	0	5	5	10	1	6

Table A5: Habitat volume fraction statistics using GMM-MRF clustering.

Habitat	LF				No LF			
	Mean (%)	SD (%)	Median (%)	IQR (%)	Mean (%)	SD (%)	Median (%)	IQR (%)
1	9	10	6	10	8	8	6	12
2	15	10	11	9	15	12	13	15
3	6	9	2	10	7	12	2	9
4	3	5	1	4	6	7	5	8
5	2	3	2	2	3	6	1	4
6	2	4	0	2	3	5	1	3
7	2	2	1	2	5	7	2	7
8	0	0	0	1	3	7	0	2
9	4	2	5	3	7	10	1	12
10	6	5	7	8	4	5	2	5
11	11	13	9	15	4	6	1	6
12	8	10	2	12	5	7	3	6
13	6	7	3	7	8	7	6	9
14	1	2	1	1	4	6	1	6
15	19	15	14	15	11	11	10	15
16	3	5	0	5	7	12	2	12

Table A6: Habitat volume fraction statistics using Spatial GMM-MRF clustering.

B Co-Selection Habitat Network

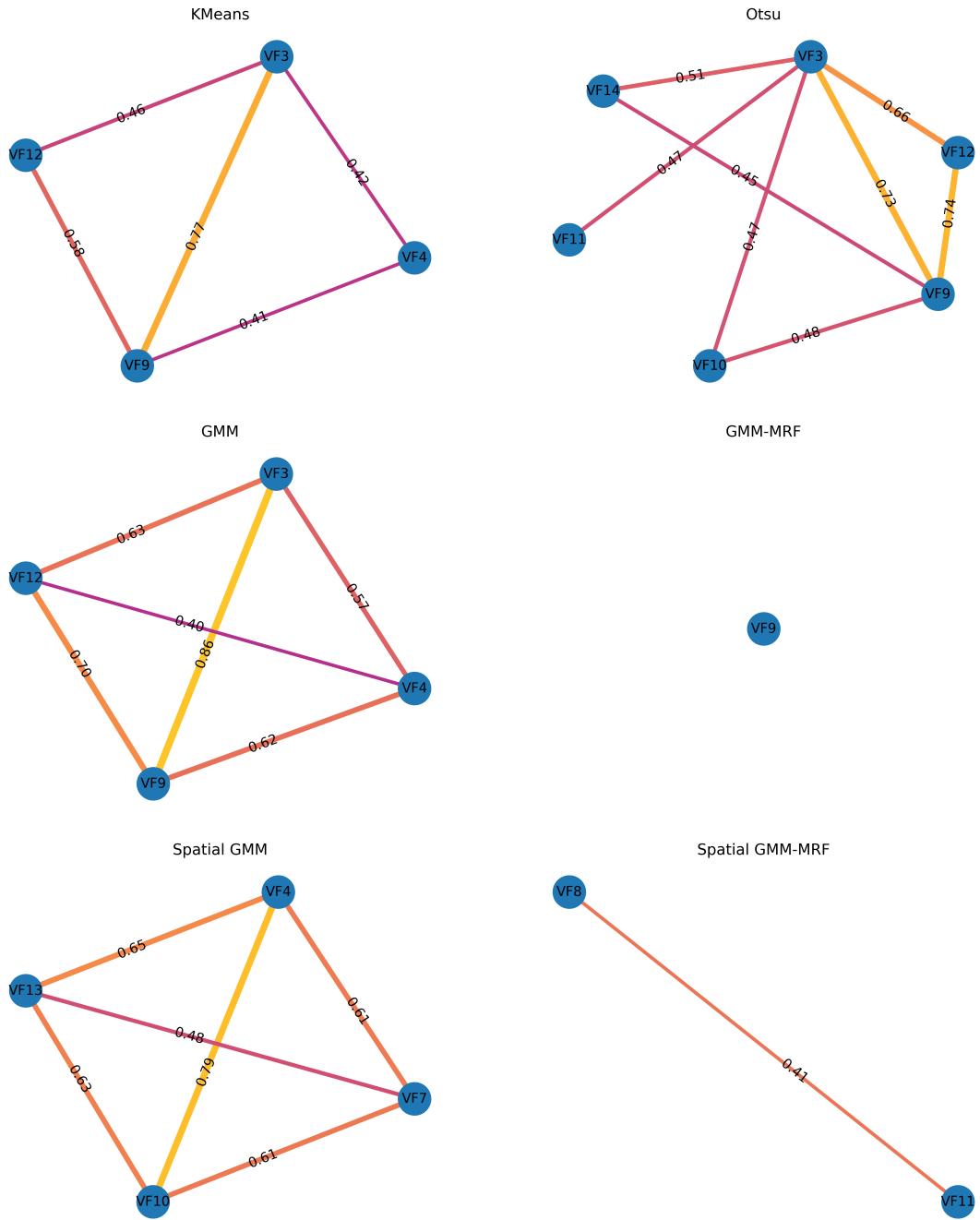


Figure B1: Co-selection frequency network graphs for the different clustering methods based on habitat volume fractions.

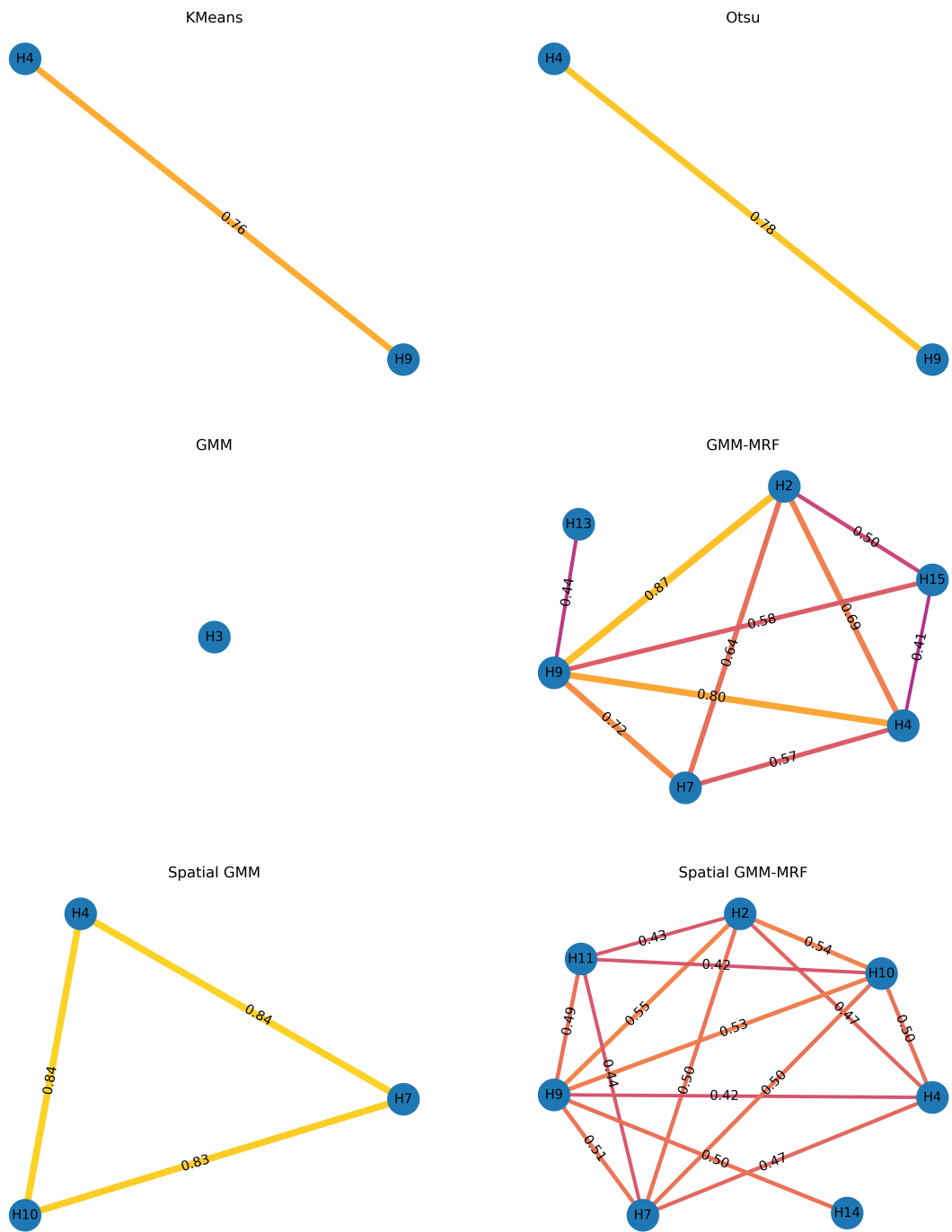


Figure B2: Co-selection frequency network graphs for the different clustering methods based on habitat presence.